Yicheng (Albert) Zhan

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EDUCATION

Ph.D. candidate (3rd year) — **Jan. 2024 - Now**
University College London - Computational Light Laboratory — **London, United Kingdom**

- **Core Fields**: Computer Vision, Computer Graphics, Computational Imaging, Augmented Reality.
- **Thesis Title**: Advanced Algorithms for Next-Generation AR and Computational Displays. (**Mentor**: Assoc. Prof. Kaan Akşit)

M.Sc. in Computer Graphics, Vision and Imaging — **Sep. 2021 - Sep. 2022**
University College London — **London, United Kingdom**

- **Graduate Result**: First Class Distinction
- **Dissertation Title**: Urban Semantic Understanding. (**Supervisor**: Assoc. Prof. Melinos Averkiou)

B.Sc. (Hons) in Software Engineering and Computer Science — **Sep. 2018 - Sep. 2021**
King's College London — **London, United Kingdom**

- **Graduate Result**: First Class Honors
- **Dissertation**: Slackbot Security Evaluation. (**Supervisor**: Prof. Jose Such)

WORK EXPERIENCE

Research Scientist Intern - Generative Model for AR display — **Mar. 2026 - Oct. 2026**
Meta Reality Labs (Supervisors: Dr. Yijing Fu and Dr. Changwon Jang) — **Redmond, United States**
Fine-tuned a flow-matching JiT foundation model for novel light phenomena synthesis, with the Display Systems Research team.

Research Scientist Intern - 3D Reconstruction — **Apr. 2025 - Sep. 2025**
Huawei Technologies Research & Development (UK) Ltd (Supervisor: Dr. Arthur Moreau) — **London, United Kingdom**
Feed-forward 3D reconstruction with Gaussian splatting and VGGT.

Research Assistant - Crime Linkage Analysis — **Jul. 2023 - Jan. 2025**
Imperial College London (Supervisor: Assoc. Prof. Dalal Alrajeh) — **London, United Kingdom**
Developed a Siamese network for serial crime linkage analysis; first-author paper published at AAAI 2026.

Research Assistant - Computer Vision — **Apr. 2023 - Dec. 2023**
University of Leeds — **Leeds, United Kingdom**

PUBLICATIONS

[1] **Yicheng Zhan**, Xiangjun Gao, Long Quan, and Kaan Akşit. *Complex-Valued 2D Gaussian Representation for Computer-Generated Holography*. **ECCV 2026** (Web)

[2] **Yicheng Zhan**, Fahim Ahmed, Amy Burrell, Matthew J. Tonkin, Sarah Galambos, Jessica Woodhams, and Dalal Alrajeh. *Enhancing Binary Encoded Crime Linkage Analysis Using Siamese Network*. **AAAI 2026** (Web)

[3] **Yicheng Zhan**, Dong-Ha Shin, Seung-Hwan Baek, and Kaan Akşit. *Complex-Valued Holographic Radiance Fields*. **ACM Transactions on Graphics** *(Presented at SIGGRAPH 2026 and highlighted at Graphics4Science Workshop)* (Web)

[4] **Yicheng Zhan**, Liang Shi, Wojciech Matusik, Qi Sun, and Kaan Akşit. *Configurable Holography: Towards Display and Scene Adaptation*. **ArXiv Preprint 2024** *(In preparation)* (Web)

[5] **Yicheng Zhan**, Koray Kavaklı, Hakan Urey, Qi Sun, and Kaan Akşit. *AutoColor: Learned Light Power Control for Multi-Color Holograms*. **SPIE VR/AR/MR 2024** (Web)

[6] Haolong Wang, **Yicheng Zhan**, Kaan Akşit, and Simeng Qiu. *Hologram Representation via Quadratic Phase Gaussian Splatting*. **ACM SIGGRAPH ASIA 2026 Tech Comm.** (Web)

[7] Runze Cheng, **Yicheng Zhan**, Josef Spjut, and Kaan Akşit. *Clustered Codebook Quantization for 2D Gaussian-based Image Compression*. **ACM SIGGRAPH 2026 Posters** *(Invited to Technical Paper Talk Panel)* (Web)

[8] Xiaoyue Fan, **Yicheng Zhan**, Amrita Mazumdar, and Kaan Akşit. *Compressing Double-Phase Holograms using 2D Gaussians*. **Eurographics 2026 Posters** (Web)

[9] Zicong Peng, **Yicheng Zhan**, Josef Spjut, and Kaan Akşit. *Assessing Learned Models for Phase-only Hologram Compression*. **ACM SIGGRAPH 2025 Posters** (Web)

[10] Chuanjun Zheng, **Yicheng Zhan**, Liang Shi, Ozan Cakmakci and Kaan Akşit. *Focal Surface Holographic Light Transport using Learned Spatially Adaptive Convolutions*. **ACM SIGGRAPH ASIA 2024 Tech Comm.** (Web)

SKILLS

Programming: Python, C++, CUDA, JavaScript.
Tools and Agents: Claude Code, Codex, Linux, Git, LaTeX, Manim.
Interests: Compose Hiphop Music, Sleep, Long-distance Running.
Frameworks: PyTorch, Odak, Mitsuba 3, OpenCV, NumPy.
Language skills: English (fluent), Chinese (native).